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Cutting tool

Abstract

A cutting tool (**100**) includes a top blade (**102**) and a bottom blade (**104**). The bottom blade (**104**) is pivotally coupled to the top blade (**102**). The top blade (**102**) and the bottom blade (**104**) together perform a cutting action on an object place between them. A first handle (**108**) is operatively coupled with the top blade (**102**). The first handle (**108**) defines a front portion (**109**) and a back portion (**111**). A second handle (**110**) is operatively coupled with the bottom blade (**104**). The cutting tool (**100**) is characterized in that in closed position of the cutting tool (**100**), at least a part of the second handle (**110**) at least partially encloses the back portion (**111**) of the first handle (**108**).

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1587718	12/1925	Gentile	30/247	B26B 13/26
1806555	12/1930	Gonsett	56/294	B26B 15/00
3350780	12/1966	Artru	81/358	A01G 3/033
5353504	12/1993	Pai	N/A	N/A
5685080	12/1996	Amano	30/DIG.1	B27B 17/08
5735353	12/1997	Undin	81/367	B25B 7/12
7441337	12/2007	Linden	30/239	A01G 3/0251
8434234	12/2012	Chen	30/247	A01G 3/037
8893390	12/2013	Nie	30/247	B26B 15/00
9955627	12/2017	Nakano	N/A	A01G 3/062
10406718	12/2018	Arthur	N/A	B24B 55/052
2002/0116824	12/2001	Herrmann	30/260	B26B 15/00
2005/0160606	12/2004	Yao	30/228	B26B 15/00
2008/0256805	12/2007	Maddison	30/295	A01G 3/08
2009/0151170	12/2008	Lee et al.	N/A	N/A
2011/0016730	12/2010	Wu	N/A	N/A
2012/0011729	12/2011	Kim	30/228	A01G 3/037
2016/0021819	12/2015	Nakano	30/276	A01G 3/062
2018/0325035	12/2017	Chung	N/A	F16H 3/66
2020/0346335	12/2019	Yang	N/A	B25F 3/00

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
207630076	12/2017	CN	N/A
207858895	12/2017	CN	N/A

20205106	12/2001	DE	A01G 3/037
102016211974	12/2017	DE	N/A
1574125	12/2004	EP	A01G 3/033
3081075	12/2015	EP	A01G 3/021
2462094	12/1980	FR	A01G 3/033
2008067630	12/2007	JP	N/A
20180205126	12/2017	WO	N/A

OTHER PUBLICATIONS

DE-20205106-U1 English translation; Jun. 27, 2002 DE; Meier, Johann A01G3/037. cited by examiner

International Search Report and Written Opinion of International Application No.

PCT/EP2020/083460 mailed Jan. 29, 2021, all pages cited in its entirety. cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a cutting tool. More specifically, the present disclosure relates to the cutting tool which allows easy and ergonomic handling for common users.

BACKGROUND

(2) Cutting tools such as secateurs, loppers, rakes, pliers are used for performing cutting action using one or more blades. Put simply, such tools provide increased mechanical assistance and ease of operation to a common user to conveniently perform operations such as with branches of varying dimensions. But application of conventional cutting tools may involve hardships due to the need for a user to frequently apply cutting force. Further, the cutting tools provide benefits such as trapping bypass cutting or simply “bypass” allows pruning/cutting by the cutting tools while holding an object (i.e. branch) with one of the cutting edges.

(3) However, there may be many instances where working with the cutting tools become inconvenient for a user. For instance, the cutting tools may be bulky or ergonomically less suitable which makes their application inconvenient for the common user.

(4) An example of a cutting tool is provided in German patent application 10,2016,211,974 (hereinafter referred to as '974 reference). The '974 reference provides a cutting tool. The cutting tool includes a first and a second cutting element which can be moved relative to one another. The cutting tool further includes a first and a second gripping element which can be moved relative to one another. The cutting tool includes a drive unit which in at least one operating state is provided to at least support a movement of the second cutting element relative to the first cutting element via a drive force transmission element. The drive force transmission element has a gear unit having a self-switching coupling unit. It is proposed that the drive unit, the gear unit, the coupling unit and the drive element of the drive force transmission element are arranged in this order, on or in the first handle member. The battery supplying power to the motor is located in the second handle member and is transmitted via a wire. The resultant design results in a bulky and ergonomically inconvenient cutting tool.

(5) Thus, there is a need of an improved cutting tool with an ergonomic and compact profile which allows ease of operation for different needs of users.

SUMMARY

(6) In view of the above, it is an objective of the present invention to solve or at least reduce the drawbacks discussed above. The objective is at least partially achieved by a cutting tool. A cutting tool includes a top blade and a bottom blade. The bottom blade is pivotally coupled to the top blade. Further, the top blade and the bottom blade together perform a cutting action on an object placed between them. The cutting tool further includes a first handle and a second handle. The first handle is operatively coupled with the top blade. Moreover, the first handle defines a front portion and a back portion. The second handle is operatively coupled with the bottom blade. The cutting tool is characterized in that, in closed position of the cutting tool, at least a part of the second handle at least partially encloses the back portion of the first handle. Thus, the present disclosure provides a simple, compact and user-friendly cutting tool which allows ease of handling for common users, by moving some of the parts of its electric drive unit into a section of the handle that is normally not touched by the user when operating the cutting tool. Advantageously the partly enclosure of the back of the first handle by a part of the second handle in closed position also protects the cutting tool from obstacles unintentionally moving into the space between the two handles. This in particular helps when maneuvering in bushes and trees. Further the movement of some of the parts of its electric drive unit into a section of the handle that is normally not touched by the user allows the placement of the users hand nearby the pivot point of blades and handles. By this even users with small hands can easily operate the cutting tool.

(7) According to an embodiment of the present invention, a battery of a drive unit is housed within the second handle. This helps in reducing the dimensions of the first handle so that the first handle remains even more partially enclosed within the second handle.

(8) According to an embodiment of the present invention, the drive unit includes a motor, a gearbox, a drum, a printed circuit board assembly (PCBA) and a battery to selectively provide the supplemental motor force. This supplemental motor force by the drive unit provides the mechanical advantage to assist cutting action by the blades. Additionally, the drive unit may be enclosed within the second handle. Thus, the size of the first handle can be further reduced.

(9) According to an embodiment of the present invention, the drive unit is operatively coupled to at least one of the top blade and the bottom blade. Further, the drive unit is configured to selectively provide a supplemental motor force to assist a movement of at least one of the top blade and the bottom blade to perform the cutting action. This makes the cutting tool of the present disclosure applicable and suitable as per varying requirement (i.e. cutting assistance or mechanical advantage) of different users.

(10) According to an embodiment of the present invention, the cutting tool includes a switch to activate or deactivate the drive unit. This allows to have a safe and convenient operation of the cutting tool with the provision of the switch.

(11) According to an embodiment of the present invention, the second handle may comprise a recess in which the first handle is arrangeable or arranged in closed position of the cutting tool. Additionally, a wall of the second handle may form the recess and may have a contour, preferably an arc-shaped or curved contour, corresponding to a contour, preferably an arc-shaped or curved contour, of the first handle such that the corresponding contours of first and second handle may be arrangeable or arranged closely adjacent to each other in a closed position of the cutting tool. In such way, the first handle and the second handle are enabled to be located very close to each other in a closed state of the cutting tool. Further, the handles can be formed ergonomically.

(12) According to an embodiment of the present invention, the second handle may define a longitudinal axis, in particular corresponding to an axis of a drive shaft of the drive unit, wherein the motor and/or the gear box and/or the drum and/or a bearing for supporting the drive shaft may be arranged consecutively relative to the longitudinal axis and wherein the battery and/or the printed circuit board assembly may be arranged laterally offset to the motor and/or the gear box and/or the drum and/or a bearing for supporting the drive shaft relative to the longitudinal axis.

Such an arrangement of the components of the drive unit within the handle enables a very balanced tool, since the masses of the components are distributed in an optimized way.

(13) According to an embodiment of the present invention, at least the second handle may comprise two half-shells forming a cavity in an assembled state of the second handle, in particular wherein the drive unit may be arrangeable or arranged in the cavity, wherein at least one of the half-shells may comprise one or two or more reinforcement structures, in particular reinforcement and support ribs or reinforcement and support walls, which are configured to support components of the drive unit arrangeable or arranged in the cavity and to reinforce the outer handle wall of the second handle against deformation caused during cutting action. In such way, assembling of the drive unit within the handle is simplified. The handle as defining a hollowing structure is stiffened and made resistant against mechanical deformation. At the same time, advantageous support for the distinct components of the drive unit is provided.

(14) According to an embodiment of the present invention, a first half-shell may comprise a plurality of connection domes, in particular thread domes, which structurally reinforce the outer wall of the first handle, in particular wherein four connection domes are arranged to define a quadrilateral plane in which at least one component of the drive unit, in particular a sub-unit including a gear box and/or a drum or a sub-unit including a motor and/or a battery and/or a printed circuit board assembly, may be arranged. Thus, an advantageous reinforcement structure is created by integrating the thread domes that are used for screw connections to assembly the half-shells of the handle to each other.

(15) According to an embodiment of the present invention, the second handle comprises at least one end portion and at least one central portion, wherein the end portion defines a first width $W_{sub.1}$ and the central portion defines a second width $W_{sub.2}$ being different from the first width $W_{sub.1}$, in particular wherein $W_{sub.1}$ is at least $1.4 \times W_{sub.2}$ or at least $1.6 \times W_{sub.2}$ or at least $1.8 \times W_{sub.2}$ or at least $2 \times W_{sub.2}$. The foregoing dimensions have been shown to enable a balanced tool with good ergonomic properties.

(16) According to an embodiment of the present invention, the first handle may define a third width $W_{sub.3}$, wherein the difference of first width $W_{sub.1}$ minus the second width $W_{sub.2}$ may substantially equal the third width $W_{sub.3}$. Such a cutting tool comprises a first handle that forms a compact and closed unit with the second handle in a closed state of the cutting tool.

(17) According to an embodiment of the present invention, first and/or second and/or third width may lie within the same plane, in particular wherein the plane corresponds to the plane in which the closing movement of the first and second handle and/or of the top and bottom blade occurs. Thus, compactness can be increased and an approximately monolithic outer appearance of the cutting tool can be achieved.

(18) Other features and aspects of this invention will be apparent from the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention will be described in more detail with reference to the enclosed drawings, wherein:

(2) FIG. 1 shows a perspective view of a cutting tool, in accordance with an aspect of the present disclosure;

(3) FIG. 2 shows a perspective view of a drive unit within the cutting tool, in accordance with an aspect of the present disclosure;

(4) FIG. 3 shows a perspective view of an adapter of the cutting tool, in accordance with an aspect of the present disclosure;

- (5) FIG. 4 shows a perspective view of the cutting tool of FIG. 1 after disassembly of a removable nut, in accordance with an aspect of the present disclosure;
- (6) FIG. 5 shows a perspective view of the cutting tool having a first handle which is not enclosed by a second handle of the cutting tool, in accordance with another aspect of the present disclosure; and
- (7) FIG. 6 shows a perspective view of a drive unit within the cutting tool, similar to FIG. 2 with some more details.

DESCRIPTION OF EMBODIMENTS

(8) The present invention will be described more fully hereinafter with reference to the accompanying drawings, in which example embodiments of the invention incorporating one or more aspects of the present invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art. For example, one or more aspects of the present invention may be utilized in other embodiments and even other types of structures and/or methods. In the drawings, like numbers refer to like elements.

(9) Certain terminology is used herein for convenience only and is not to be taken as a limitation on the invention. For example, “upper”, “lower”, “front”, “rear”, “side”, “longitudinal”, “lateral”, “transverse”, “upwards”, “downwards”, “forward”, “backward”, “sideward”, “left,” “right,” “horizontal,” “vertical,” “upward”, “inner”, “outer”, “inward”, “outward”, “top”, “bottom”, “higher”, “above”, “below”, “central”, “middle”, “intermediate”, “between”, “end”, “adjacent”, “proximate”, “near”, “distal”, “remote”, “radial”, “circumferential”, or the like, merely describe the configuration shown in the Figures. Indeed, the components may be oriented in any direction and the terminology, therefore, should be understood as encompassing such variations unless specified otherwise.

(10) FIG. 1 illustrates a cutting tool **100**. The cutting tool **100** of the present disclosure is illustrated as secateurs, however, the present disclosure may be readily applied to any cutting tool **100** such as, but not limited to, a lopper, shears, scissors etc. with one, two or more cutting edges. The cutting tool **100** includes one or more blades i.e. a top blade **102** and a bottom blade **104**. Preferably the top blade **102** is a replaceable blade. The cutting tool **100** further includes a nut **106**. Preferably this nut **106** is removable. The nut **106** is provided at the pivoted position of the top blade **102** and the bottom blade **104**.

(11) In some embodiments, the removable nut **106** may be any one or more of a coupling nut, a flange nut, a hex nut, a lock nut, a slotted nut, a square nut, a wheel nut and the like.

(12) The bottom blade **104** is pivotally coupled to the top blade **102**. The top blade **102** and the bottom blade **104** together perform a cutting action on an object (generally a branch) placed between them. In one embodiment the cutting tool **100** further includes a lid **132** That allows an easy replacement of the battery. Preferably the cutting tool is using a replaceable battery or even can use different ones out of a set of replaceable batteries.

(13) As illustrated in FIG. 1, the cutting tool **100** includes one or more handles i.e. a first handle **108** and a second handle **110**. The first handle **108** is operatively coupled with the top blade **102**. The first handle **108** defines a front portion **109** and a back portion **111**. The second handle **110** is operatively coupled with the bottom blade **104**. In closed position of the cutting tool **100**, at least a part of the second handle **110** at least partially encloses the back portion **111** of the first handle **108**. This improves ergonomic profile of the combination of the first handle **108**, and the second handle **110**, particularly in the closed position of the cutting tool **100**.

(14) In some embodiments, any or both of the top blade **102** and the bottom blade **104** may be one or more of a clip point blade, a drop point blade, a gut hook blade, a hawkbell blade, a needle point blade, a straight back blade, a sheepsfoot blade, a spear point blade or any other blade as used or known in the art.

(15) In some embodiment, the top blade **102** and the bottom blade **104** may be made of aluminum or any other material as used or known in the art. Choice of the material for any of the top blade **102** and the bottom blade **104** will generally depend upon factors such as life, application, weight or other parameters related to the cutting tool **100**. However, the present disclosure is not to be limited by the type of material, type, or specification of the top blade **102** and the bottom blade **104** in any manner.

(16) During implementation, at least one of the top blade **102** and the bottom blade **104** may be a fixed blade while the other performs desired cutting action. Alternately or additionally, both the top blade **102** and the bottom blade **104** may be movable during the cutting action.

(17) In some embodiments, the coupling of the one or more blades i.e. the top blade **102** and the bottom blade **104** to its respective handles **108** and **110** may be done by gluing, tying, screwing and the like.

(18) FIG. 2 illustrates the cutting tool **100** which includes a drive unit **130**. The drive unit **130** includes a motor **112** and a gearbox **114**. The drive unit **130** further includes a drum **116** comprising a clutch, a retraction spring and a winch. Moreover, the drive unit **130** includes a battery **120** a printed circuit board assembly (PCBA) **118**. In the depicted embodiment the battery **120** is covered by the lid **132**. The lid **132** may be assembled or disassembled using any arrangement such as, but not limited to, sliding connection, snap-fit connection or push-fit connection to change the battery **120**. The drive unit **130** may be operatively coupled to at least one of the top blade **102** and the bottom blade **104**. The drive unit **130** is configured to selectively provide a supplemental motor force to assist a movement of at least one of the top blade **102** and the bottom blade **104** to perform the cutting action. In the illustrated embodiments, the drive unit **130** is housed in the second handle **110**. Further, the battery **120** of the drive unit **130** is housed within the second handle **110**. This helps in reducing the dimensions of the first handle **108**, particularly the combined profile of the second handle **110** enclosing the first handle **108** becomes more ergonomic and easier to handle for common users.

(19) In some embodiments, the battery **120** may be a primary battery, or a secondary battery. For example, the battery **120** may be one or more of a lithium-ion, a nickel cadmium, a nickel-metal hydride, lead-acid.

(20) In a preferred embodiment, all components of the drive unit **130** are housed in a single handle i.e. the second handle **110** as best illustrated in FIG. 2 with the cutting tool being in closed status. This helps in reducing the thickness, dimensions, ergonomic profile or spatial footprint of the first handle **108**. This even allows the combined profile of the second handle **110**, and the first handle **108** to be looked at as one part, particularly during closed state of the first handle **108**, when at least a part of the second handle **110** at least partially encloses the back portion **111** of the first handle **108**. This advantageous ergonomic profile can be seen with the side view of the cutting tool **100** of FIG. 2. By this it also becomes possible to realize the first handle **108** as a slim bottom handle that allows the operation of the cutting tool even with small hands.

(21) In some embodiments, outer boundary lines (not shown) of both the first handle **108** and the second handle **110** may be parallel to each other. Alternately or additionally, the outer boundary lines of the one or more of handles i.e. the first handle **108** and the second handle **110** may be non-parallel or non-aligned with respect to each other.

(22) In an embodiment, the cutting tool **100** includes a slider or trigger to activate or deactivate the drive unit **130**. In the depicted embodiment this slider or trigger is realized in top of the cutting tool **100** as slider or trigger **122**, alternatively it as well could be realized as a slider or trigger. On engaging, moving or pressing the slider or trigger **122** the drive unit **130** gets switched on and off and thus the cutting tool **100** becomes convenient and safe to handle by common users. The drive unit **130** may be deactivated for instances which may require to save/extend battery life or allow uninterrupted operation of the power-assisted cutting tool **100**, such as for low battery conditions. In such instances of the deactivated drive unit **130**, it could be possible that the motor **112** will

remain in a disengaged/OFF state leading to a purely user-based cutting action by the cutting tool **100**, without any supplemental motor force. In one embodiment the slider or trigger **122** might also look the cutting tool **100** in a closed status.

(23) In some embodiments, the slider or trigger **122** may be a button, a knob, a single pole single throw switch, a single pole double throw switch, a double pole single throw switch, a double pole double throw switch, a push-button switch, a toggle switch, a limit switch, a float switch, and the like. Alternately or additionally, a button, a knob, or any other switching means as used or known in the relevant art may be present in place of or additionally to the slider or trigger **122**.

(24) FIG. 3 illustrates details with the first handle **108** (shown in the bottom of the drawing) and the second handle **110** (shown on the top of the drawing) of the cutting tool **100** in a closed position. The cutting tool **100** having a spring **124**. The cutting tool **100** further includes a rope **126** transmitting the force from the drive unit **130** inside the second handle **110** into the first handle **108** and an adapter **128** inside of the first handle **108**. The adapter **128** finds application for pre-fixing the spring **124** and the rope **126** with the first handle **110**, without any direct engagement or contact of the spring **124** and the rope **126** with the first handle **110**, as best illustrated in FIG. 4. The top blade **102** passes through the adapter **128**. The piercing of the top blade **102** through the adapter **128** helps in securing or locking of the top blade **102** within the adapter **128**, and/or the bottom handle **110** as per the requirement.

(25) During implementation, a connection of the adapter **128** with the top blade **102** may be achieved by pushing, sliding, and piercing the top blade **102** through the adapter **128**. The adapter **128** may be made up of one or more of a material like plastic, rubber, polymers, glass and the like. The adapter **128** allows smooth assembly or disassembly during movement of the top blade **102** (through the adapter **128**) by simple pulling action on the top blade **102**, after removal of the removal nut **106** as illustrated in FIG. 4.

(26) FIG. 4 illustrates the cutting tool **100** post disassembly of the removable nut **106**. This sets up the cutting tool **100** for removal or disassembly of the top blade **102** from the cutting tool **100**. The removable nut **106** may be disassembled by use of hand tools or by any other way convenient for the common user. Then, the disassembled top blade **102** allows its substitution with another top blade **102** (not shown), which may have a sharper edge or any other desired feature. During implementation, the top blade **102** may be substituted with another top blade **102**. The substituted blade i.e. another top blade **102**, for the top blade **102**, may have different design, profile etc. with same basic structure which may allow it to desirably engage with the adapter **128**.

(27) FIG. 5 illustrates the cutting tool **100** with the first handle **108** disengaged and not enclosed by the second handle **110**. The first handle **108** and the second handle **110** have one or more inner sides **115**, **117** respectively. In some embodiments, the diameter of both the first handle **108**, and the second handle **110** may be advantageously rounded. Any or both of the first handle **108**, and the second handle **110** may have a soft grip surface to increase the ease of use or to avoid slipping during application. Advantageously, an undesirable crushing, in particular a pinching of a skin of an operator's hand, trapping of the operator's hand between the first handle **108** and the second handle **110**, and the like may be avoided by the elastic, rounded design of the one or more inner sides **115**, **117** of both the first handle **108** and the second handle **110**.

(28) Further, as illustrated in FIG. 5, a recess **134** is formed by the second handle **110**. The form of the recess **134** corresponds with the form of the first handle **108** such that the first handle **108** can be arranged within the recess closely to the second handle **110**. The recess **134** and the first handle **108** comprise corresponding contours that are arc-shaped or curved. The recess is delimited by an end portion **142** and a central portion **144** of the second handle **110**, wherein the end portion has a width $W_{sub.1}$ being substantially larger than a width $W_{sub.2}$ of the central portion, such that the first handle **108**, having a width $W_{sub.3}$ fits within the recess **134** to such an extent that the first handle **108** forms a closed outer contour with the second handle **110** and does not project beyond the contour of the second handle **110** at least within a plane defined by the plane drawn up by the

first handle **108**, the second handle **110** and a pivot point, e.g. defined by the nut **106**.

(29) FIG. **6** illustrates the cutting tool **100** in a closed state. The first handle **108** is accommodated within the recess **134**. The drive unit **130** is arranged completely within a cavity formed by a first half-shell of the second handle **110**. A drive shaft (not referenced) of the drive unit **130** defines an axis A. Motor **112**, gear box **114**, drum **116** and a bearing **148** for supporting the drive shaft are arranged consecutively or in other words in a row. In this specific embodiment, the motor shaft and the gear shaft coupled to the drive shaft extend within one axis A, however, the skilled person may appreciate that there may be further possibilities of arrangement. The printed circuit board assembly PCBA and the battery **120** are not arranged within a row with the other components of the drive unit **130**, but are laterally offset to them. In particular, the longitudinal axis of the printed circuit board assembly PCBA and the battery **120** is offset laterally to the axis A and may be arranged parallel or non-parallel thereto.

(30) Furthermore, the second handle **108** comprises two shells forming a hollow cavity for accommodation of the drive unit **130**, wherein one half-shell is removed in the view of FIG. **6**. The half-shell that is still visible comprises thread domes **138** for receiving screws for connecting both half-shells to each other. It shall be appreciated that the thread domes **138** further increase the resistance of the second handle **110** against mechanical deformation, by the way of their form as support posts. This effect is further increased by arranging the thread domes **138** at specific positions. Four thread domes **134** are arranged to define a quadrilateral plane in which a sub-unit including a gear box **114** and a drum **116** or a sub-unit including a motor **112**, a battery **120** and a printed circuit board assembly PCBA is arranged. Additionally or alternatively, it is possible that a shaft **140** of the bottom blade **104** may be coupled to the second handle **110** in a region close to thread domes **134**, e.g. in such way that the shaft **140** protrudes in the direction of the axis A, wherein, e.g. two thread domes **134** form a support and/or fixation and/or a coupling recess for the shaft **140**. Additionally, the second handle **110** may comprise reinforcement structures **136** such as reinforcement and support ribs or reinforcement and support walls in order to support the components of the drive unit **130** fixedly within the cavity of the second handle **110** and to provide the second handle **110** with structural reinforcements to resist mechanical deformations.

(31) In the drawings and specification, there have been disclosed preferred embodiments and examples of the invention and, although specific terms are employed, they are used in a generic and descriptive sense only and not for the purpose of limitation of the scope of the invention being set forth in the following claims.

LIST OF ELEMENTS

(32) **100** Cutting Tool **102** Top Blade **104** Bottom Blade **106** Removable Nut **108** First handle **109** Front Portion **110** Second handle **111** Back Portion **112** Motor **114** Gear Box **115** Inner Side **116** Drum **117** Inner Side **118** Printed Circuit Board Assembly (PCBA) **120** Battery **122** Slider or Trigger **124** Spring **126** Rope **128** Adapter **130** Drive unit **132** Lid **134** Recess **136** Reinforcement structures **138** Thread domes **140** Shaft **142** End portion **144** Central portion **146** Front portion **148** Bearing A Axis of drive shaft W.sub.1 First width W.sub.2 Second width W.sub.3 Third width

Claims

1. A cutting tool comprising: a top blade; a bottom blade pivotally coupled to the top blade, wherein the top blade and the bottom blade together perform a cutting action on an object placed therebetween; a first handle operatively coupled with the top blade, wherein the first handle comprises a front portion, a back portion and a first inner side; and a second handle operatively coupled with the bottom blade; wherein, in a closed position of the cutting tool, at least a part of the second handle at least partially encloses the back portion of the first handle, and wherein the cutting tool further comprises a drive unit and a slider or trigger to activate and deactivate the drive unit, wherein the drive unit is configured to selectively provide a supplementary motor force to

assist pivoting the top blade relative to the bottom blade, wherein the slider or trigger is disposed at the second handle and operable by a user of the cutting tool to power on and power off the drive unit and to selectively provide the supplementary motor force via the drive unit, wherein the second handle comprises a recess defined by a second inner side of the second handle, wherein the first inner side and the second inner side are both arcuate and the first handle is conformal with the second handle in the closed position, and wherein the second handle comprises the recess in which the first handle is arrangeable or arranged in the closed position of the cutting tool, wherein a wall of the second handle forming the recess has a contour of an arc-shaped or curved contour, corresponding to a contour of an arc-shaped or curved contour, of the first handle such that the corresponding contours of first and second handle are arrangeable or arranged closely adjacent to each other in the closed position of the cutting tool.

2. The cutting tool of claim 1, wherein a battery of the drive unit is housed within the second handle.

3. The cutting tool of claim 2, wherein the drive unit comprising a motor, a gearbox, a drum, Printed Circuit Board Assembly (PCBA) and the battery to selectively provide the supplementary motor force, wherein the drive unit is enclosed within the second handle.

4. The cutting tool of claim 2, wherein the drive unit is operatively coupled to at least one of the top blade and the bottom blade.

5. The cutting tool of claim 1, wherein at least the second handle comprises two half-shells forming a cavity in an assembled state of the second handle, wherein the drive unit is arrangeable or arranged in the cavity, wherein at least one of the half-shells comprises one or two or more reinforcement structures in a form of reinforcement and support ribs or reinforcement and support walls, which are configured to support components of the drive unit arrangeable or arranged in the cavity and to reinforce the outer handle wall of the second handle against deformation caused during cutting action.

6. The cutting tool of claim 5, wherein a first half-shell comprises a plurality of connection domes including thread domes, which structurally reinforce the outer wall of the second handle, wherein four connection domes are arranged to define a quadrilateral plane in which at least one component of the drive unit comprising a sub-unit including a gear box or a drum or a sub-unit including a motor or a battery or a printed circuit board assembly (PCBA), is arranged.

7. The cutting tool of claim 1, wherein the second handle comprises at least one end portion and at least one central portion, wherein the at least one end portion defines a first width and the at least one central portion defines a second width being different from the first width, wherein the first width is at least 1.4 times the second width.

8. The cutting tool of claim 7, wherein the first handle defines a third width, wherein a difference of the first width minus the second width substantially equals the third width.

9. The cutting tool of claim 8, wherein the first width, the second width or the third width lie within the same plane, wherein the same plane corresponds to a plane in which a closing movement of the first and second handle or of the top and bottom blade occurs.

10. The cutting tool of claim 1, wherein a length of the first handle extending from the front portion to the back portion is between 30% and 70% of a length of the second handle.

11. The cutting tool of claim 1, wherein the first handle is operably coupled to the second handle via an adapter disposed within the first handle, and wherein the adapter is operably coupled to the top blade such that the top blade extends through the adapter within the first handle.

12. A cutting tool comprising: a top blade; a bottom blade pivotally coupled to the top blade, wherein the top blade and the bottom blade together perform a cutting action on an object placed therebetween; a first handle operatively coupled with the top blade, wherein the first handle defines a front portion and a back portion; and a second handle operatively coupled with the bottom blade; wherein, in a closed position of the cutting tool, at least a part of the second handle at least partially encloses the back portion of the first handle, wherein a drive unit disposed within the second

handle comprises a motor, a gearbox, a drum, Printed Circuit Board Assembly (PCBA) and a battery to selectively provide supplemental motor force, wherein the drive unit is enclosed within the second handle, wherein the second handle defines a longitudinal axis corresponding to an axis of a drive shaft of the drive unit, wherein the motor, the gear box, the drum, and a bearing for supporting the drive shaft are all disposed coaxially along the longitudinal axis, wherein the battery or the printed circuit board assembly is arranged laterally offset to the motor or the gear box or the drum or the bearing for supporting the drive shaft relative to the longitudinal axis, and wherein the cutting tool further comprises a slider or trigger to activate and deactivate the drive unit, wherein the drive unit is configured to selectively provide a supplementary motor force to assist a movement of at least one of the top blade and the bottom blade, and wherein the slider or trigger is disposed at the second handle and operable by a user of the cutting tool to power on and power off the drive unit and to selectively provide the supplementary motor force via the drive unit.

13. The cutting tool of claim 12, wherein the first handle is operably coupled to the second handle via an adapter disposed within the first handle, and wherein the adapter is operably coupled to the top blade such that the top blade extends through the adapter within the first handle.
